

4(a)	<i>any three from lung cancer to max 2</i> (tar) is a carcinogen ; acts as mutagen / causes mutations / described ; I mutant cells e.g. causes changes to sequence of, bases / nucleotides, in DNA detail ; I uncontrolled mitosis e.g. tumour suppressor genes switched off protooncogenes to oncogenes / oncogenes switched on (cell cycle) checkpoints fail to function AW cell moves from, G0 / inactive state of cell cycle, to G1 phase (for mitosis) no programmed death / AW <i>chronic bronchitis to max 2</i> (tar) paralyses / destroys / damages / AW, cilia ; I weakens cilia A damage to <u>ciliated</u> (epithelial) cells consequence to <u>cilia</u> action ; <i>must be related to mp1</i> e.g. <u>cilia</u> do not, waft away / AW, mucus loss of synchronous rhythm of <u>cilia</u> (tar causes), goblet cells / mucous glands, to, increase in size / enlarge / AW ; A (tar / accumulation of mucus, causes) inflammation more / excess, mucus produced (by, goblet / mucous gland cells) ; I mucus accumulates	3
4(b)	<i>correct units for volume cm³ ;</i> <i>SA <u>and</u> volume calculation (SA) = 54 (V) = 27 ;</i>	2
4(c)	A → B → C ; A 6:1 → 3:1 → 2:1 A 1 → 2 → 3	1

Question	Answer	Marks
4(d)	<i>any two from (in emphysema)</i> alveolar walls break down / alveoli burst, to, produce one large air sac / give a lower SA:V or diseased, tissue / areas / alveoli, has lower SA:V (than healthy tissue) ; removed tissue has (higher proportion of), large air sacs / diseased alveoli ; remaining tissue has, healthy / undamaged / functional / only, alveoli ; AW A volume reduced and functional surface area of lung remains the same A remaining tissue has higher functional surface area (than before surgery)	2
4(e)	<i>any two from</i> 1 <u>only</u> oxygenated blood passes through left side of heart ora or oxygenated blood in left side and deoxygenated blood in right side of heart ; 2 septum in heart separates (oxygenated and deoxygenated) blood ; 3 oxygenated blood is in pulmonary, venules / veins <u>and</u> deoxygenated blood is in, pulmonary arteries / arterioles ; 4 (because mixing prevented by) double circulation / double circulatory system / pulmonary and systemic circulations ;	2